

# Executive Summary: Thesis Progress & Meeting Objectives

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**Date:** 27 November 2025

**Status:** Technical experiments complete; scope freeze needed

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## Methodology Briefing

### The Problem

**"Given a bracket or housing and a load case, where should 2-6 screws go to maximize stiffness (minimize compliance)?"**

Engineers currently solve this by rules of thumb and manual FEA. This thesis automates it using two modern approaches.

### Why Dual Approach?

- Literature compares: PINN vs FEA, MMC vs SIMP
- **Gap:** No one has compared **MMC vs PINN specifically for discrete fasteners**
- **This thesis fills that gap** with direct empirical comparison

### Approach A: PINN (Physics-Informed Neural Network)

#### Core Idea:

- Treat FEA solver as differentiable → run gradient-based optimizer → log (screw positions, compliance)
- Train neural network surrogate on that data to approximate compliance function

#### Why "Physics-Informed":

- Inputs encode geometry and load case
- Training data comes from physics-accurate differentiable FEA
- Network learns the same structural behavior as FEA

#### Key Phrase:

"PINN turns expensive FEA into a fast differentiable black box. There is an upfront training cost (2 hours), but after that, optimization and design studies are almost free."

#### Benefits:

- After training: predicts compliance in microseconds (0.009 ms)
- Matches FEA within  $\leq 1.1\%$  across multiple geometries and loads
- Makes repeated queries  $10^3\times$  faster

### Approach B: MMC (Moving Morphable Components)

#### Core Idea:

- Represent each screw as explicit geometric component (circular inclusion in 2D, cylindrical in 3D)
- Optimize component parameters (positions, radii) directly using topology optimization

### Mechanism:

- Density/projection field converts components into element stiffness values
- Standard FEA + sensitivity analysis drives optimizer (MMA/SLSQP) to move components

### Key Phrase:

"MMC is a purist's method: everything is governed by FEA and optimization theory, no learning. It's transparent and precise, but you pay for that every iteration (118 ms/iter)."

### Benefits:

- No training data or neural nets needed
- Explicit, manufacturable geometry by construction
- Can enforce spacing and edge margins directly

### Costs:

- Every iteration runs full FEA (118 ms/iter)
- Must re-optimize from scratch for each new load case or geometry

### What the Thesis Compares

1. **Convergence Speed:** PINN (high upfront, then free) vs MMC (zero upfront, but expensive per iteration)
  2. **Solution Quality:** Do both find similar optimal layouts?
  3. **Generalization:** PINN (one-shot across geometries) vs MMC (re-optimize each time)
  4. **Implementation:** Complexity, scalability, applicability domains
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## Current Results Snapshot

### Multi-Geometry PINN (Approach A)

- **Dataset:** 1020 samples across 3 geometries (L-bracket, tapered plate, ribbed channel)
- **Accuracy:** MAE  $\leq 12.4$  J,  $\leq 1.1\%$  MAPE on all high-energy loads
- **Speed:** 0.009 ms/inference ( $\geq 10^3\times$  faster than FEA at 10.4 ms)
- **Generalization:** One-shot inference across 3 geometries / 5 load cases
- **Training:** 2 hours one-time cost (1500 epochs)

### MMC (Approach B)

- **Geometries:** L-bracket, tapered plate, ribbed channel (all completed)
- **Performance:** 118 ms/iteration, 30 iterations per run
- **Status:** Operational on all three geometries with convergence logs
- **Note:** Compliances in normalized units; conversion to Joules pending

### Comparative Summary

| Method   | Time/Iteration | Training Cost  | Generalization           |
|----------|----------------|----------------|--------------------------|
| Diff-FEA | 10.4 ms        | None           | N/A                      |
| PINN     | 0.009 ms       | 2 h (one-time) | Cross-geometry           |
| MMC      | 118 ms         | None           | Per-load re-optimization |

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## Proposed Final Tasks Checklist

### Technical Work:

- ☐ Execute Ansys validation on 3-5 selected layouts (target: <20% error)
- ☐ Normalize MMC compliances to Joules and regenerate unified comparison plots
- ☐ Produce dataset coverage figure for 1020-sample corpus
- ☐ Minor cleanup: fix ribbed-channel shear error if time permits

### Writing Work:

- ☐ Draft Chapters 1-3 (Introduction, Literature Review, Theoretical Framework)
  - ☐ Draft Chapters 4-5 (Methodology, Results)
  - ☐ Draft Chapters 6-8 (Comparative Analysis, Discussion, Conclusion)
  - ☐ Incorporate supervisor feedback
  - ☐ Final formatting and proofreading
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## Key Decisions Needed

### 1. Scope Freeze

- ☐ **Is current 2D/2.5D work sufficient?** (L-bracket + tapered + ribbed)
- ☐ **Is 3D benchmark required?** If yes:
  - ☐ PINN-only 3D acceptable? (MMC remains 2D-only)
  - ☐ Which geometry: extruded L-bracket or simplified motor-housing?

### 2. Ansys Validation

- ☐ **How many layouts?** (Suggested: 3-5)
- ☐ **Error threshold acceptable?** (Suggested: <20%)
- ☐ **Which layouts?** (Best diff-FEA, best PINN, best MMC per geometry)

### 3. MMC Normalization

- ☐ **Must be done before thesis?** Or acceptable to explain scaling in text?
- ☐ **Timeline:** Before writing or during writing phase?

### 4. Writing Timeline

- ☐ **Chapter delivery schedule:**
  - Ch. 1-3 draft: \_\_\_\_\_ weeks
  - Ch. 4-5 draft: \_\_\_\_\_ weeks
  - Full draft: \_\_\_\_\_ weeks
- ☐ **Review rounds:** How many? (Suggested: 2-3)
- ☐ **Latest full-draft date:** \_\_\_\_\_ (back-solve from defense date)

### 5. Thesis Balance

- ☐ **Emphasis:** More on comparative discussion/generalization vs. deep derivations vs. more experiments?
  - ☐ **MMC ribbed-channel values:** Acceptable if explained, or must fix first?
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## Approved Thesis Scope Statement

One sentence description:

### Example options:

- "Comparative study of multi-geometry PINN vs MMC on three 2D benchmark assemblies with Ansys validation on selected layouts"
  - "Comparative study of multi-geometry PINN vs MMC on three 2D benchmark assemblies, with 3D PINN extension, and Ansys validation on selected layouts"
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## Supervisor Notes

Decisions made:

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Additional requirements:

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Next meeting date: \_\_\_\_\_

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## Key Figures to Reference

- Multi-geometry PINN performance: data/results/comprehensive\_results\_table.md
  - Unified speed comparison: data/results/unified\_speed\_comparison.png
  - Unified compliance comparison: data/results/unified\_compliance\_comparison.png
  - MMC convergence plots: data/results/mmc\*\_compliance.png
  - Full results table: data/results/comprehensive\_results\_table.md
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**Next Steps:** Fill in decisions during meeting, then freeze scope and proceed with final tasks.